

Week 8 (Thursday, August 14)

Support Vector Machines

CMPE-257: Machine Learning - Summer 2025



SJSU

**SAN JOSÉ STATE
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Today's plan...

Main Topic: Support Vector Machines

5:30-7:00 Session 1

7:00-7:30 Break

7:30-9:20 Session 2

9:20-9:30 Term Projects and HW8

Session 1:

- Maximal Margin Classifier and Support Vectors

Session 2:

- Support Vector Machines
- Hands-on coding session 8

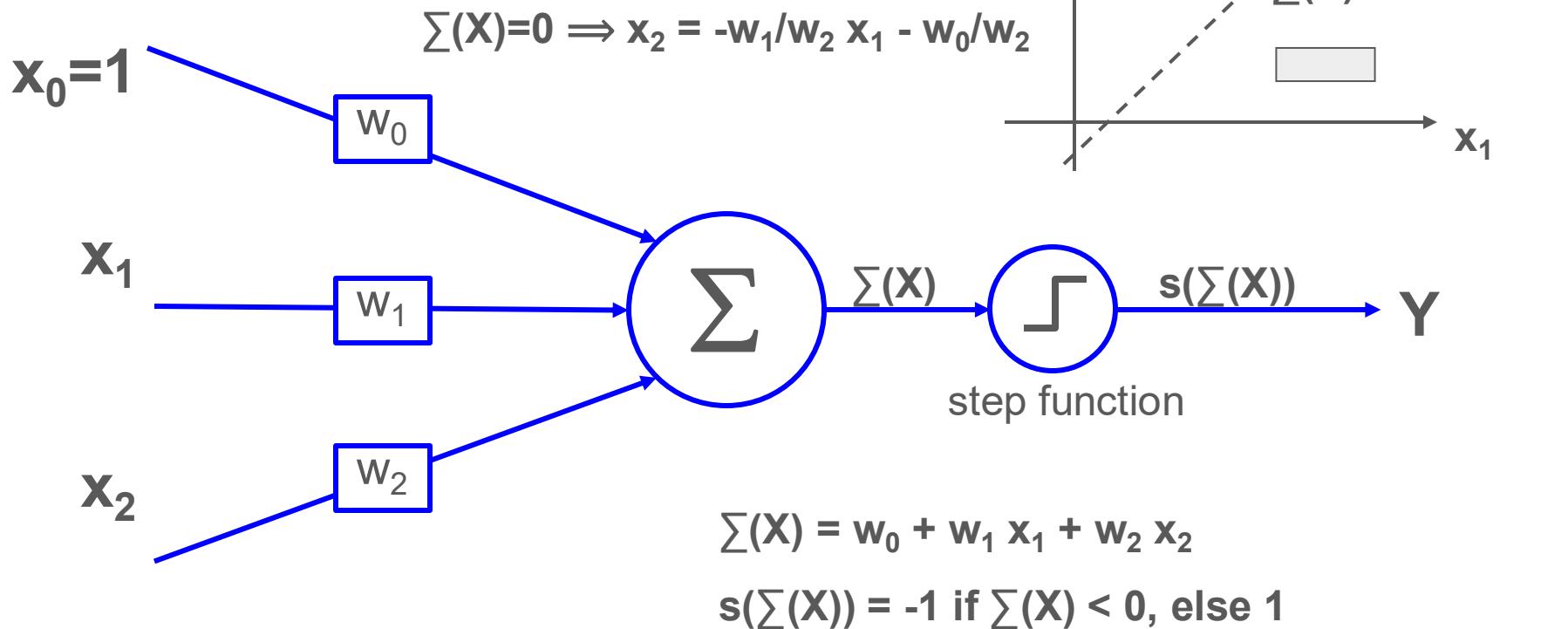
Seventh Homework is due on Sunday!

Week 8 - Session 1

Maximal Margin and Support Vectors

- Maximal Margin Classifier
- Support Vector based Classification
- Support Vector based Regression

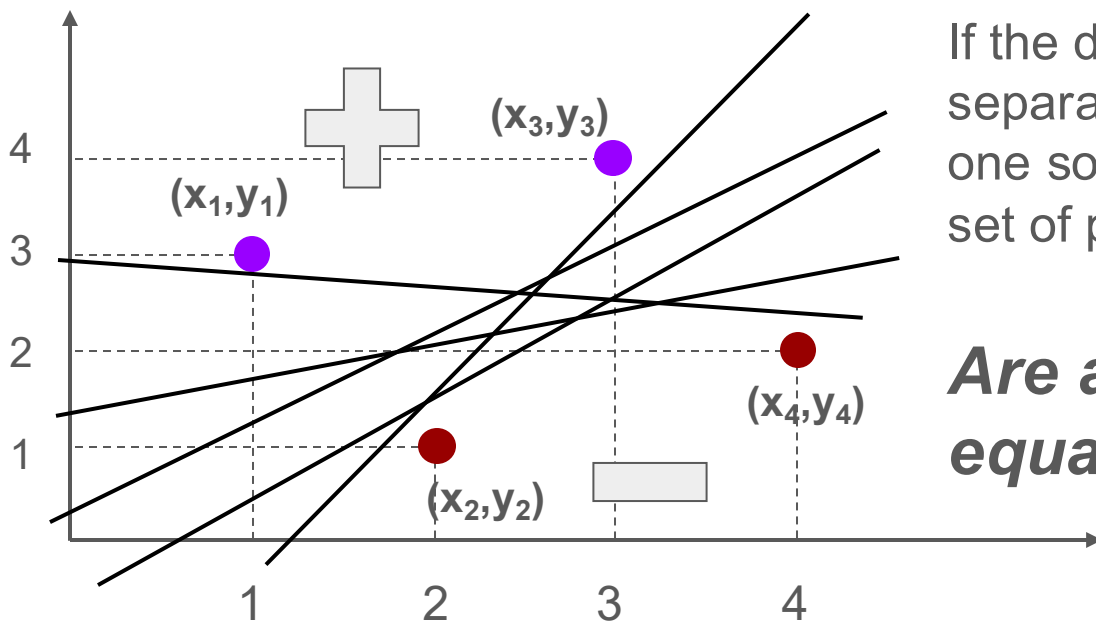
The perceptron as a binary classifier





PLA: perceptron learning algorithm

Updates the weights sequentially using the error signal...



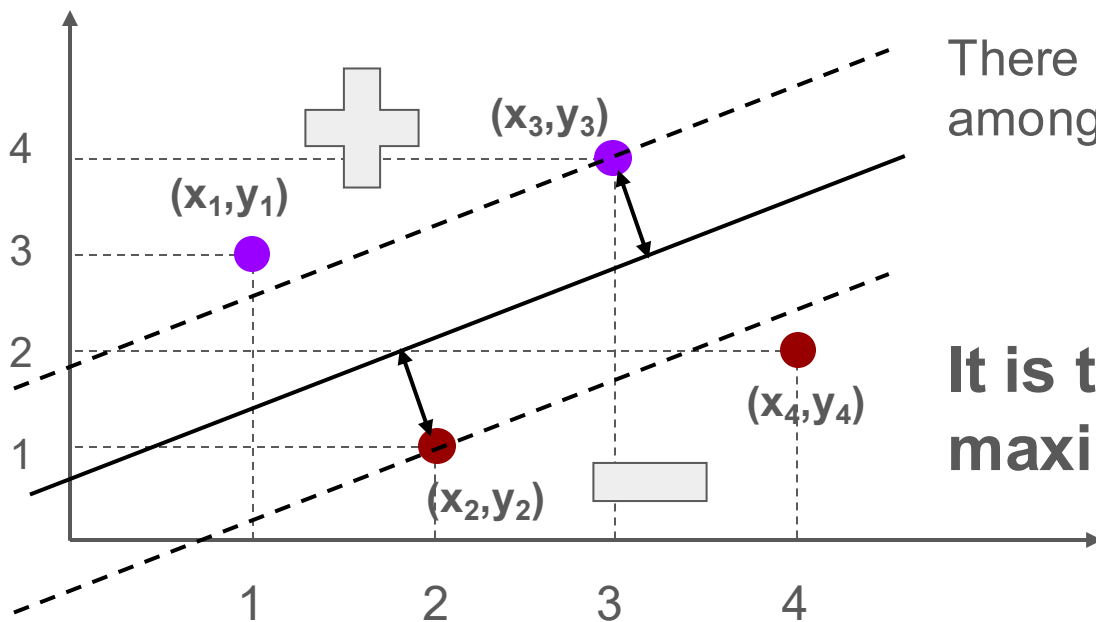
If the data points are linearly separable, the PLA will find one solution (from an infinite set of possible solutions!)

Are all such solutions equally good? NO



Maximal margin classifier

The separating line farthest from the training observations...

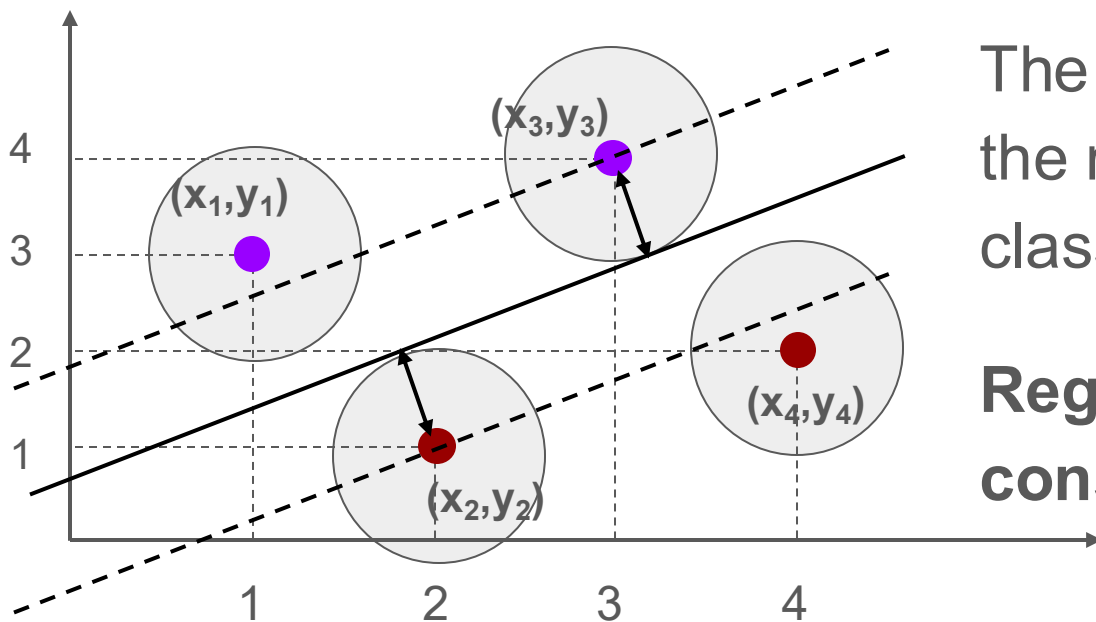


There is one best solution among them...

It is the one that maximizes the margin!

Maximal margin classifier and noise

The separating line farthest from the training observations...

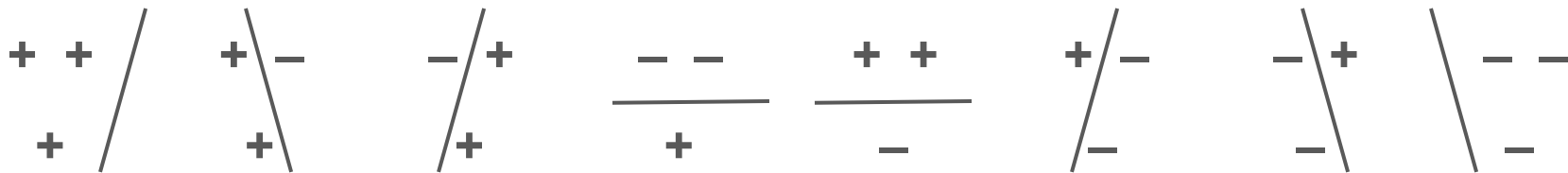


The largest the margin,
the most data noise the
classifier will tolerate!

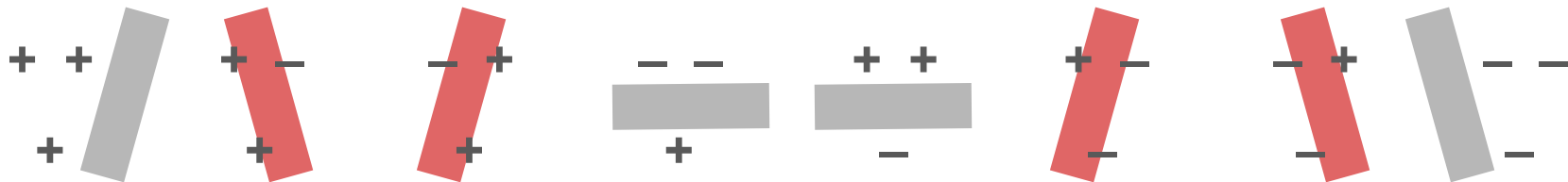
**Regularization by
construction!**

From our previous discussion on VC dimension...

- For a very narrow margin: $N = 3 \Rightarrow m_G(3) = 8$



- For a wide margin: $N = 3 \Rightarrow m_G(3) \leq 8$



- Wider margins reduce the effective VC dimension of the model!

Maximal margin classifier: observations

- The margin is the closest distance from a data sample to the separating line
- The separating line with maximal margin is actually the optimal separating line (i.e. the most immune to noise!)
- This concept is extensible to more dimensions:
 - optimal separating plane for $\mathbb{R}^3$
 - optimal separating hyperplane for $\mathbb{R}^N$ with $N > 3$

Maximal margin classifier: more observations

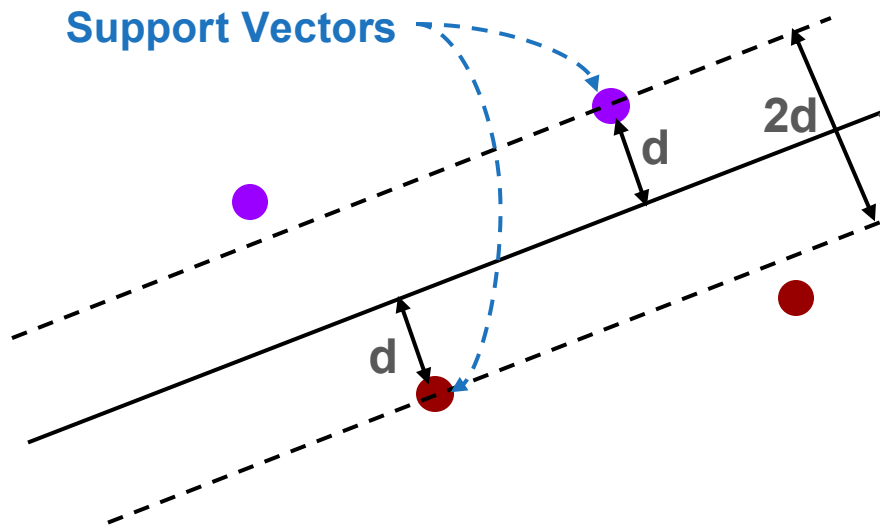
- The optimal separating hyperplane depends only on a few train data samples: the support vectors
- The support vectors do not only determine the optimal separating hyperplane but the size of the margin too
- The larger the margin the better the classifier!
- The optimal separating hyperplane can be derived from a training dataset by using QP (quadratic programming)

Finding the maximal margin hyperplane

- For a given a set of training observations $\{ ([x_{n1}, x_{n2}, \dots x_{nK}], y_n) \}$ with $n = 1, 2, \dots N$; $k = 1, 2, \dots K$; $y_n = \{-1, 1\}$, and
- a hyperplane of the form $w_0 + \sum_k w_k x_k = 0 \Rightarrow w_0 + \mathbf{w}^T \mathbf{x} = 0$ that separates the data
- It can be seen that: $y_n (w_0 + \mathbf{w}^T \mathbf{x}_n) > 0$ for all training data, and
- $| w_0 + \mathbf{w}^T \mathbf{x} | / || \mathbf{w} ||$ is the shortest distance between a point $\mathbf{x}$ and the hyperplane, which
- can be rewritten as $y_n (w_0 + \mathbf{w}^T \mathbf{x}_n) / || \mathbf{w} ||$ because $y_n = \{-1, 1\}$

Finding the maximal margin hyperplane

- We want to find the set of coefficients $(w_0, \mathbf{w})$ that optimizes the following expression $\operatorname{argmax}_{(w_0, \mathbf{w})} (\min_{n=1, \dots, N} y_n (w_0 + \mathbf{w}^T \mathbf{x}_n) / \|\mathbf{w}\|)$



Find the hyperplane $w_0 + \mathbf{w}^T \mathbf{x} = 0$ that (i) correctly separates the classes and (ii) maximizes the minimum possible distances from training data points in both classes

Finding the maximal margin hyperplane

- Because $\gamma (w_0 + \mathbf{w}^T \mathbf{x}) = 0$ still represents the same hyperplane for every value of γ , we can smartly select an γ as normalization factor to simplify the maths...
- Let γ be equal to $\min_n y_n (w_0 + \mathbf{w}^T \mathbf{x}_n)$, then
- the maximal margin hyperplane optimization problem becomes:
 - $\operatorname{argmax}_{(w_0, \mathbf{w})} \|\mathbf{w}\|^{-1}$ (or alternatively: $\operatorname{argmin}_{(w_0, \mathbf{w})} \frac{1}{2} \|\mathbf{w}\|^2$)
 - subject to: $y_n (w_0 + \mathbf{w}^T \mathbf{x}_n) \geq 1$ for $n = 1, 2, \dots, N$



Maximal margin hyperplane exercise

- Consider the following set of observations:
 - $\{ P_1: [1,3], +1), P_2: ([2,1], -1), P_3: ([3,5], +1), P_4: ([4,1], -1) \}$
- For which the line $-1.0 - 0.4 x_1 + 0.8 x_2 = 0$ happens to be a maximal margin line
- Compute the margin and prove that P_1 and P_2 are support vectors but P_3 and P_4 are not.

$$\text{Margin} = \|\mathbf{w}\|^{-1} = 1 / \sqrt{(-0.4)^2 + (0.8)^2} = 1 / \sqrt{0.8} = 1.118$$



Maximal margin hyperplane exercise

Shortest distance between a point $P_x: (x_1, x_2)$ and the line:

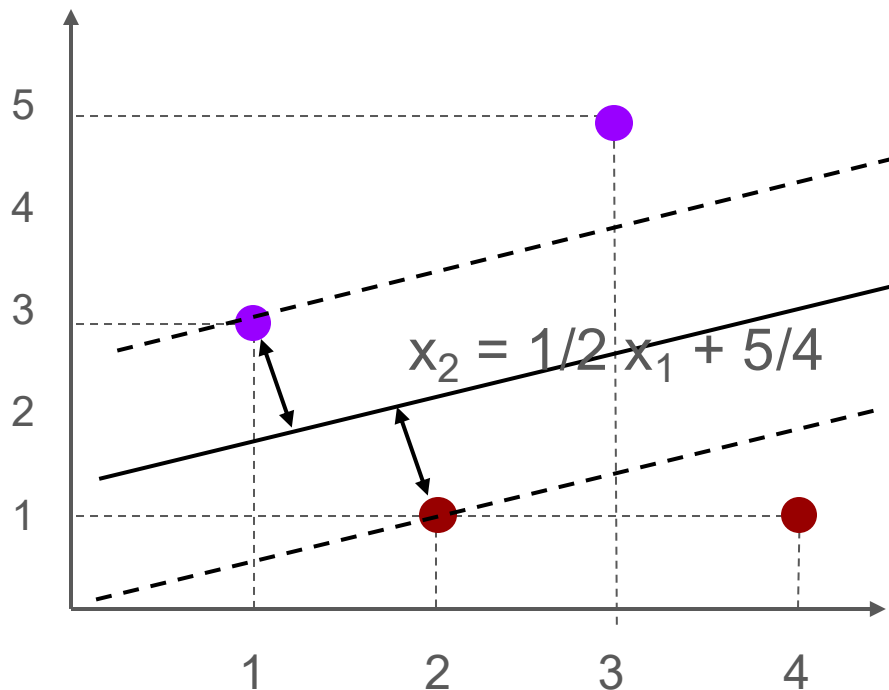
$$\underbrace{|-1.0 - 0.4x_1 + 0.8x_2|}_{= 1 \text{ for support vectors}} \times \text{margin}$$

$$P_1: (1, 3) \Rightarrow |-1.0 - 0.4 + 2.4| = 1 \quad \checkmark$$

$$P_2: (2, 1) \Rightarrow |-1.0 - 0.8 + 0.8| = 1 \quad \checkmark$$

$$P_3: (3, 5) \Rightarrow |-1.0 - 1.2 + 4.0| = 1.8 \quad \times$$

$$P_4: (4, 1) \Rightarrow |-1.0 - 1.6 + 0.8| = 1.8 \quad \times$$





QP: quadratic programming

- $\mathbf{u}^* \leftarrow \text{QP}(\mathbf{Q}, \mathbf{p}, \mathbf{A}, \mathbf{c})$

- find $\mathbf{u}^*$ that minimize $\frac{1}{2} \mathbf{u}^T \mathbf{Q} \mathbf{u} + \mathbf{p}^T \mathbf{u}$
- subject to $\mathbf{A} \mathbf{u} \geq \mathbf{c}$

QP Solver

- In our case of interest...

- $\mathbf{u}^T = [w_0 \ w_1 \ \dots \ w_K]$ $\mathbf{p}^T = [0 \ 0 \ 0 \ \dots \ 0]$ $\mathbf{c}^T = [1 \ 1 \ 1 \ \dots \ 1]$

- $\mathbf{Q} = \begin{pmatrix} 0 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \dots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{pmatrix}$

$$\mathbf{A} = \begin{pmatrix} y_1 & y_1 x_{11} & y_1 x_{12} & \dots & y_1 x_{1K} \\ y_2 & y_2 x_{21} & y_2 x_{22} & \dots & y_2 x_{2K} \\ \vdots & \vdots & \vdots & \dots & \vdots \\ y_N & y_N x_{N1} & y_N x_{N2} & \dots & y_N x_{NK} \end{pmatrix}$$



QP formulation exercise

- Consider the following set of observations:
 - $\{ P_1: [1,3], +1), P_2: ([2,1], -1), P_3: ([3,5], +1), P_4: ([4,1], -1) \}$
- Write the maximal margin problem in the standard QP form:

$$\mathbf{u}^* \leftarrow \text{QP}(\mathbf{Q}, \mathbf{p}, \mathbf{A}, \mathbf{c})$$

find $\mathbf{u}^*$ that minimize $\frac{1}{2} \mathbf{u}^T \mathbf{Q} \mathbf{u} + \mathbf{p}^T \mathbf{u}$; subject to $\mathbf{A} \mathbf{u} \geq \mathbf{c}$

$$\mathbf{u} = \begin{pmatrix} w_0 \\ w_1 \\ w_2 \end{pmatrix} \quad \mathbf{p} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad \mathbf{c} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \quad \mathbf{Q} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \mathbf{A} = \begin{pmatrix} 1 & 1 & 3 \\ -1 & -2 & -1 \\ 1 & 3 & 5 \\ -1 & -4 & -1 \end{pmatrix}$$



Lagrange dual of the QP problem

- Given the primal form of the problem:
 - find $\mathbf{u}^*$ that minimize $\frac{1}{2} \mathbf{u}^T \mathbf{Q} \mathbf{u} + \mathbf{p}^T \mathbf{u}$
 - subject to $\mathbf{A} \mathbf{u} \geq \mathbf{c}$
- $\mathbf{u}^*$ is optimal for the primal, if and only if $(\mathbf{u}^*, \boldsymbol{\alpha}^*)$ is a solution to the dual optimization problem:
 - $\max_{\boldsymbol{\alpha} \geq 0} \min_{\mathbf{u}} L(\mathbf{u}, \boldsymbol{\alpha})$ (Lagrange Function) $\sum_n \alpha_n (\mathbf{c}_n - \mathbf{a}_n^T \mathbf{u})$
 - where $L(\mathbf{u}, \boldsymbol{\alpha}) = \frac{1}{2} \mathbf{u}^T \mathbf{Q} \mathbf{u} + \mathbf{p}^T \mathbf{u} + \underbrace{\boldsymbol{\alpha}^T (\mathbf{c} - \mathbf{A} \mathbf{u})}_{\text{Lagrange Multipliers}}$



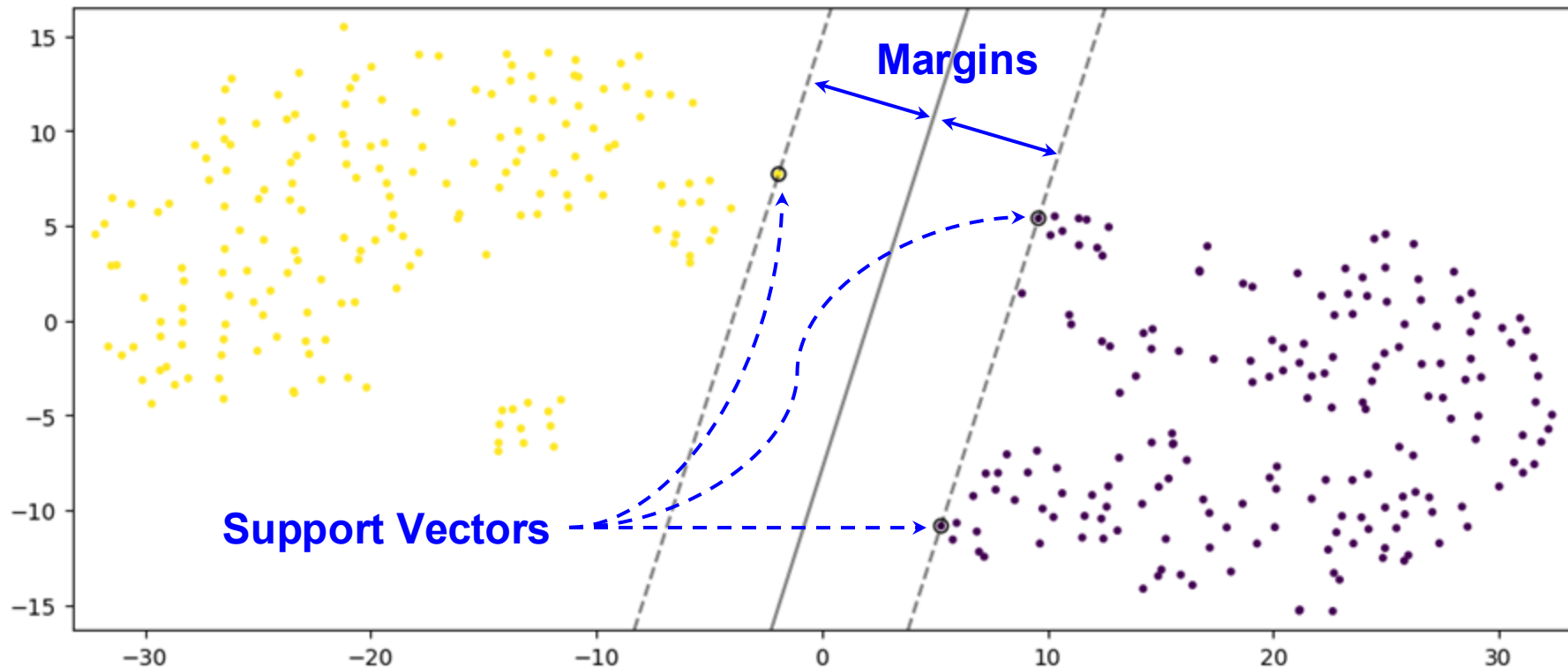
Lagrange dual formulation exercise

- Consider the following set of observations:
 - $\{ P_1: [1,3], +1), P_2: ([2,1], -1), P_3: ([3,5], +1), P_4: ([4,1], -1) \}$
- Write the maximal margin problem in its lagrangian form:

$$\max_{\alpha \geq 0} \min_{\mathbf{u}} L(\mathbf{u}, \boldsymbol{\alpha})$$

$$\begin{aligned} \text{with } L(\mathbf{u}, \boldsymbol{\alpha}) = & \frac{1}{2} (w_1^2 + w_2^2) \\ & + \alpha_1 (1 - w_0 - w_1 - 3 w_2) + \alpha_2 (1 + w_0 + 2w_1 + w_2) \\ & + \alpha_3 (1 - w_0 - 3w_1 - 5 w_2) + \alpha_4 (1 + w_0 + 4w_1 + w_2) \end{aligned}$$

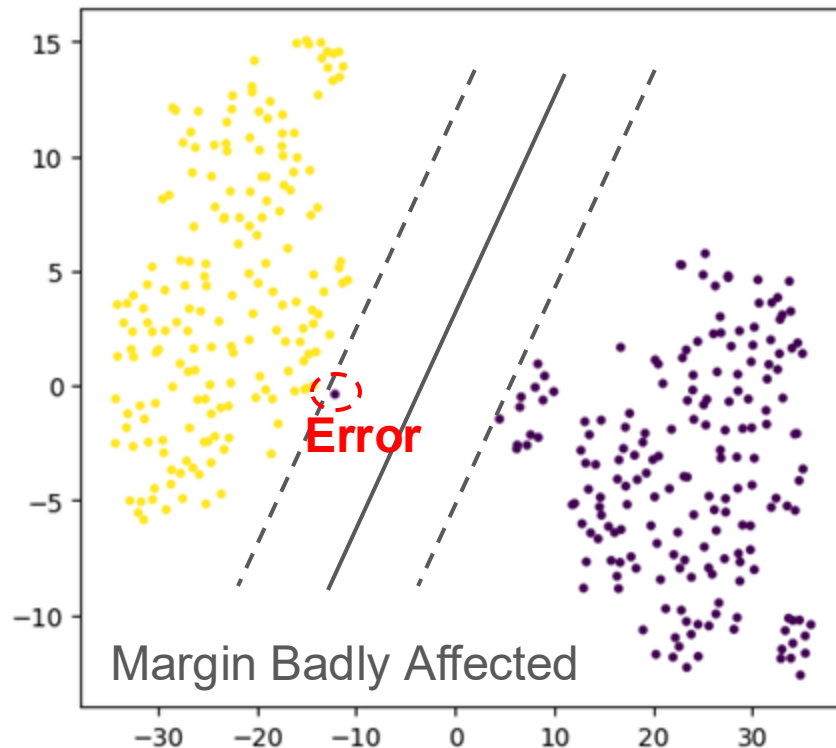
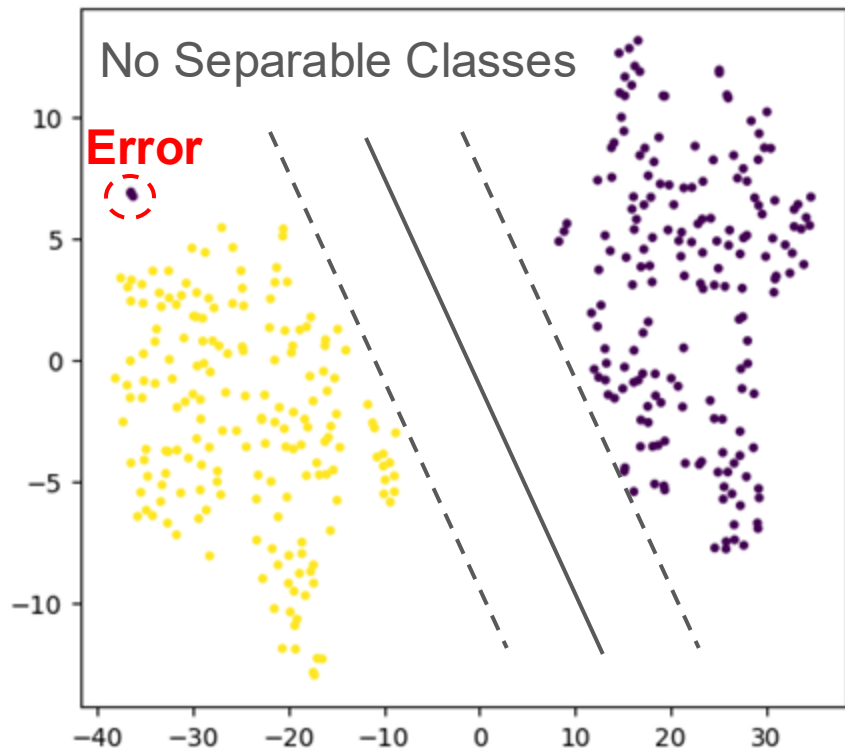
Maximal margin classifier on separable classes



Soft margin classifier (support vector classifier)

- There are cases where the maximal margin classifier might not be either feasible or the best option:
 - The data is not linearly separable
 - The margin is badly affected by some data samples
- Relax the maximal margin constraint:
 - allow some observations to be in the incorrect side of the margin and/or the incorrect side of the hyperplane

Cases where a soft margin classifier is useful

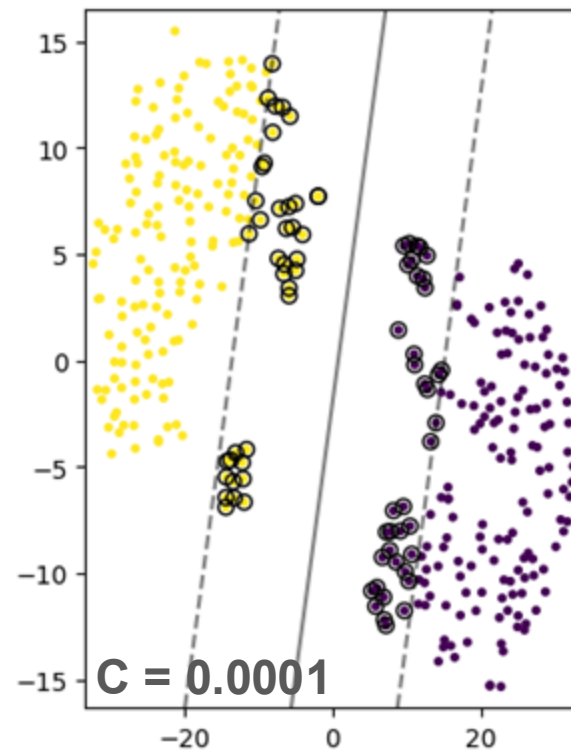
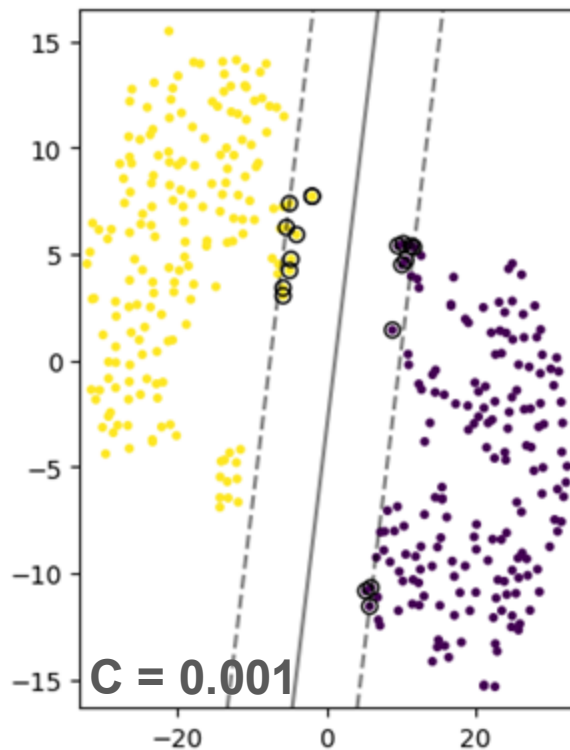
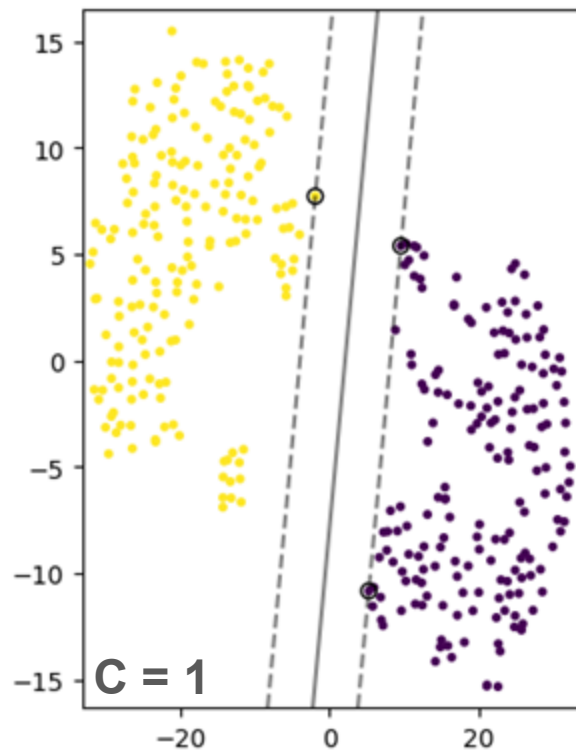




Soft margin classifier formulation

- The soft margin hyperplane optimization problem is:
 - $\operatorname{argmax}_{(w_0, \mathbf{w}, \mathbf{e})} \|\mathbf{w}\|^{-1}$ (or alternatively: $\operatorname{argmin}_{(w_0, \mathbf{w}, \mathbf{e})} \frac{1}{2} \|\mathbf{w}\|^2$)
 - subject to: $y_n (w_0 + \mathbf{w}^T \mathbf{x}_n) \geq 1 - e_n$ for $n = 1, 2, \dots, N$
 - and $e_n \geq 0$ and $\sum_n e_n \leq E$, with $E \geq 0$
- The e_n 's are slack variables that allow individual data points to be on the wrong side of the margin ($e_n > 0$) or hyperplane ($e_n > 1$)
- E is a tuning parameter that controls the softness of the margin, i.e. a budget for the amount of margin violations allowed

Soft margins with different values of C ($\sim 1/E$)

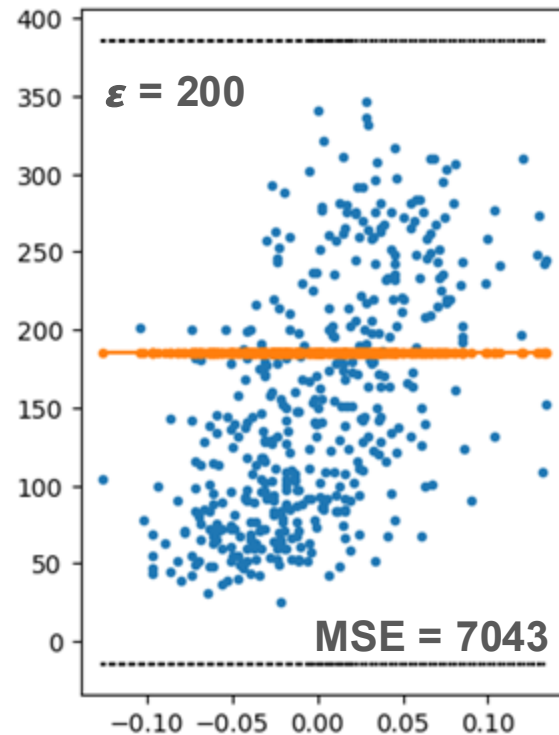
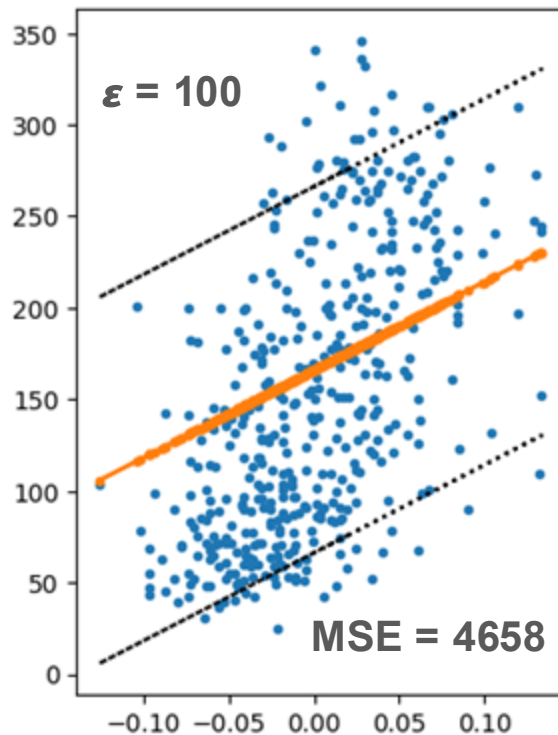
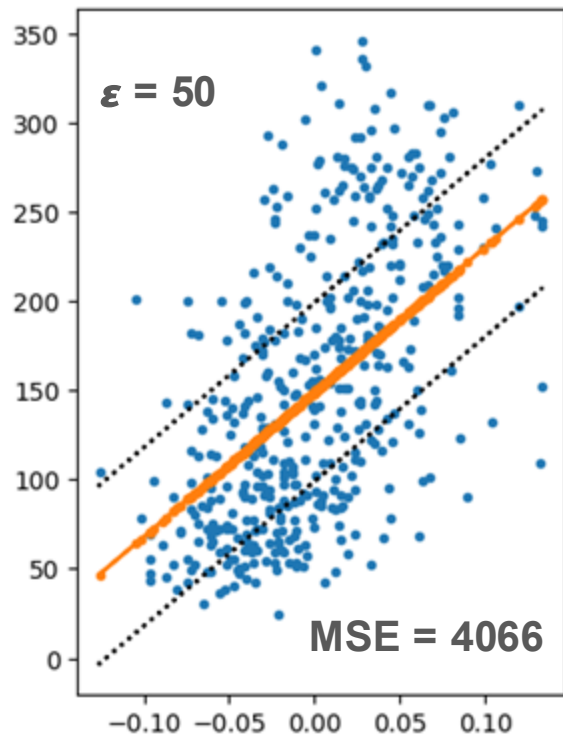




Support vector regression formulation

- A variant to the maximum margin classifier problem formulation can be used to perform a linear regression:
 - $\operatorname{argmax}_{(w_0, \mathbf{w})} \|\mathbf{w}\|^{-1}$ (or alternatively: $\operatorname{argmin}_{(w_0, \mathbf{w})} \frac{1}{2} \|\mathbf{w}\|^2$)
 - subject to: $|y_n - (w_0 + \mathbf{w}^T \mathbf{x}_n)| < \varepsilon$ for $n = 1, 2, \dots, N$
- Where $(w_0 + \mathbf{w}^T \mathbf{x}_n)$ is the prediction for data sample n
- $|y_n - (w_0 + \mathbf{w}^T \mathbf{x}_n)|$ is the absolute error of the n th prediction
- ε is a parameter defining the desired error threshold

Support vector regression example





Week 8 - Session 2

Support Vector Machines

- Kernel Methods (Linear, Polynomial, and Gaussian)
- Classification with more than two classes

Let us revisit the maximal margin and its dual problem

- The maximal margin hyperplane optimization problem was:
 - $\operatorname{argmin}_{(w_0, \mathbf{w})} \frac{1}{2} \|\mathbf{w}\|^2$
 - subject to: $y_n (w_0 + \mathbf{w}^T \mathbf{x}_n) \geq 1$ for $n = 1, 2, \dots, N$
- The dual lagrange optimization problem is:
 - $\max_{\alpha \geq 0} \min_{(w_0, \mathbf{w})} \frac{1}{2} \mathbf{w}^T \mathbf{w} + \sum_n \alpha_n (1 - y_n (w_0 + \mathbf{w}^T \mathbf{x}_n))$
 - $\Rightarrow \max_{\alpha \geq 0} \min_{(w_0, \mathbf{w})} \frac{1}{2} \mathbf{w}^T \mathbf{w} - \sum_n \alpha_n y_n \mathbf{w}^T \mathbf{x}_n - w_0 \sum_n \alpha_n y_n + \sum_n \alpha_n$

Let us optimize for $(w_0, \mathbf{w})$ first...

- $L(w_0, \mathbf{w}, \alpha) = \frac{1}{2} \mathbf{w}^T \mathbf{w} - \sum_n \alpha_n y_n \mathbf{w}^T \mathbf{x}_n - w_0 \sum_n \alpha_n y_n + \sum_n \alpha_n$
 - $\delta L(w_0, \mathbf{w}, \alpha) / \delta w_0 = 0 \Rightarrow - \sum_n \alpha_n y_n = 0 \Rightarrow \sum_n \alpha_n y_n = 0$
 - $\delta L(w_0, \mathbf{w}, \alpha) / \delta \mathbf{w} = 0 \Rightarrow \mathbf{w} - \sum_n \alpha_n y_n \mathbf{x}_n = 0 \Rightarrow \mathbf{w} = \sum_n \alpha_n y_n \mathbf{x}_n$
- Replacing back optimal w_0 and $\mathbf{w}$ into the terms in $L(w_0, \mathbf{w}, \alpha)$
 - $\frac{1}{2} \mathbf{w}^T \mathbf{w} = \frac{1}{2} \sum_n \alpha_n y_n \mathbf{x}_n^T \sum_m \alpha_m y_m \mathbf{x}_m = \frac{1}{2} \sum_n \sum_m \alpha_n \alpha_m y_n y_m \mathbf{x}_n^T \mathbf{x}_m$
 - $-\sum_n \alpha_n y_n \mathbf{w}^T \mathbf{x}_n = -\sum_n \alpha_n y_n \sum_m \alpha_m y_m \mathbf{x}_m^T \mathbf{x}_n = -\sum_n \sum_m \alpha_n \alpha_m y_n y_m \mathbf{x}_n^T \mathbf{x}_m$
 - $-w_0 \sum_n \alpha_n y_n + \sum_n \alpha_n = 0 + \sum_n \alpha_n = \sum_n \alpha_n$

We get a new optimization problem...

- Maximize the resulting Lagrangian function $L(\alpha)$
 - Minimize: $\frac{1}{2} \sum_n \sum_m \alpha_n \alpha_m y_n y_m \mathbf{x}_n^T \mathbf{x}_m - \sum_n \alpha_n$
 - Subject to: $\sum_n \alpha_n y_n = 0$ and $\alpha_n \geq 0$; for $n = 1, 2, 3 \dots N$
- The new optimization problem is only with respect to the α_n 's!
- After finding the optimal α_n^* 's
 - $\mathbf{w}^* = \sum_n \alpha_n^* y_n \mathbf{x}_n$
 - $w_0^* = y_s - \mathbf{w}^{*T} \mathbf{x}_s$ (with $\mathbf{x}_s$ being a support vector, i.e. $\alpha_s^* > 0$)

The kernel function

- The final classifier is: $g(\mathbf{x}) = \text{sign}(w_0^* + \sum_n \alpha_n^* y_n \overbrace{\mathbf{x}_n^T \mathbf{x}}^{\mathbf{w}^{*T} \mathbf{x}})$
- Which is actually: $g(\mathbf{x}) = \text{sign}(w_0^* + \sum_s \alpha_s^* y_s \mathbf{x}_s^T \mathbf{x})$
- Notice this estimator does not depends on the actual data values $\mathbf{x}$ but on their internal products with the support vectors!
- Then $g(\mathbf{x}) = \text{sign}(w_0^* + \sum_s \alpha_s^* y_s K(\mathbf{x}_s, \mathbf{x}))$, where $K(\mathbf{x}_n, \mathbf{x}_m) = \mathbf{x}_n^T \mathbf{x}_m$ is known as the linear kernel...
- In general, a kernel function is a way of combining a transform and an internal product into a more efficient computation

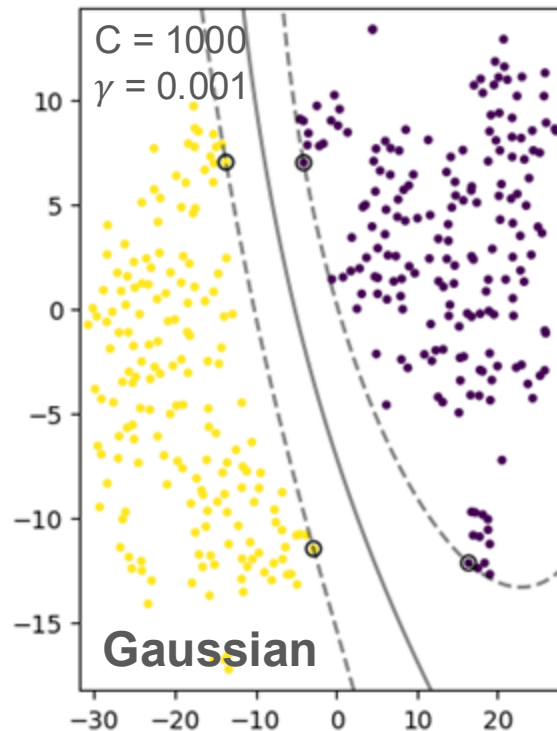
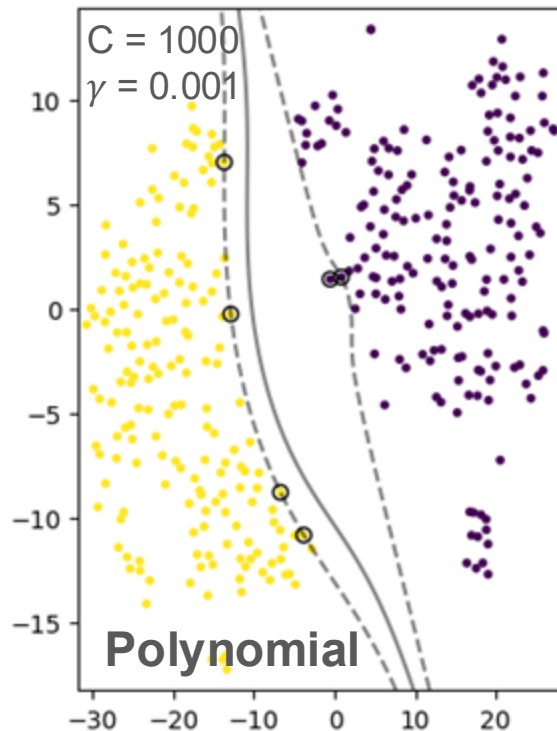
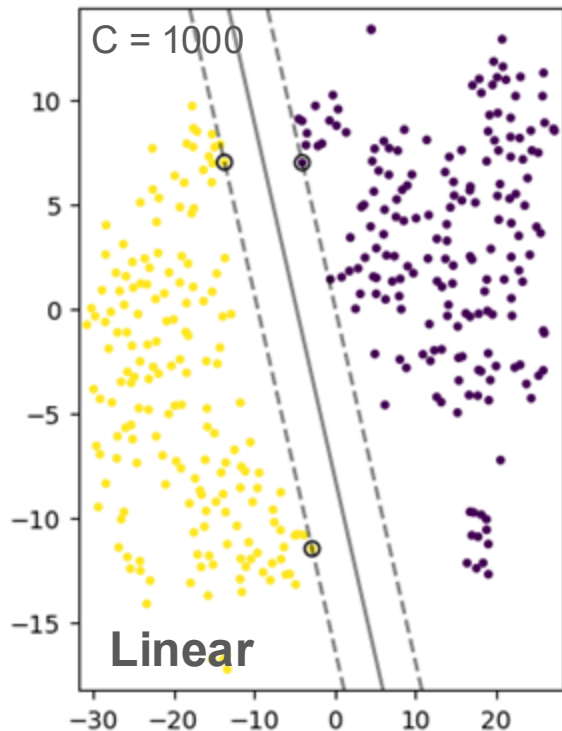


Non-linear approximations via kernel functions

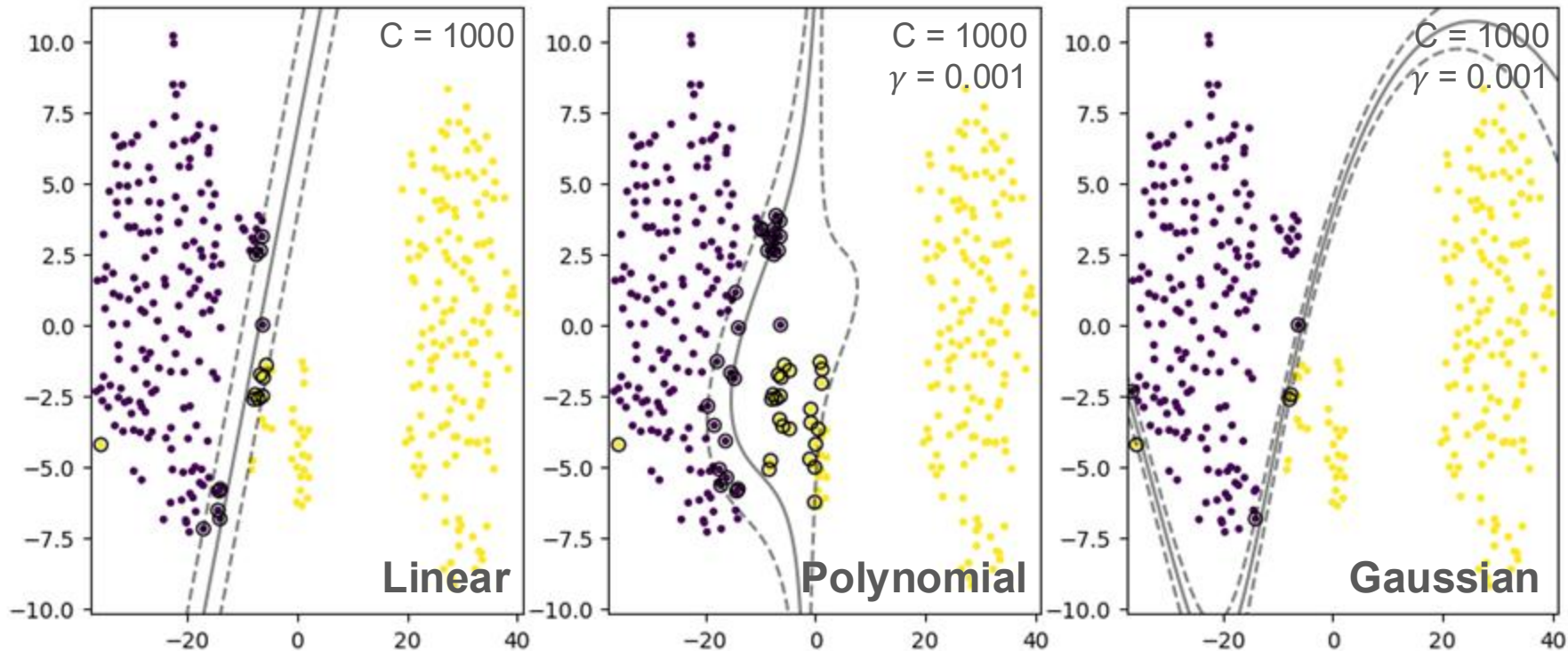
- Kernel functions can be used to augment the predictor space (similarly to how it was done with polynomial regressions!)
- Although SVM operates as a linear model in the augmented space, non-linearity is exhibited in the original problem space
- A couple of commonly used Kernel functions:
 - Polynomial kernel: $K_{\text{POLY}}(\mathbf{x}_n, \mathbf{x}) = (\gamma + (\mathbf{x}_n^T \mathbf{x}))^Q$
 - Gaussian* kernel: $K_{\text{RBF}}(\mathbf{x}_n, \mathbf{x}) = \exp(-\gamma (\mathbf{x}_n^T \mathbf{x})^2)$

* the Gaussian Kernel is also known as Radial Basis Function (RBF) Kernel

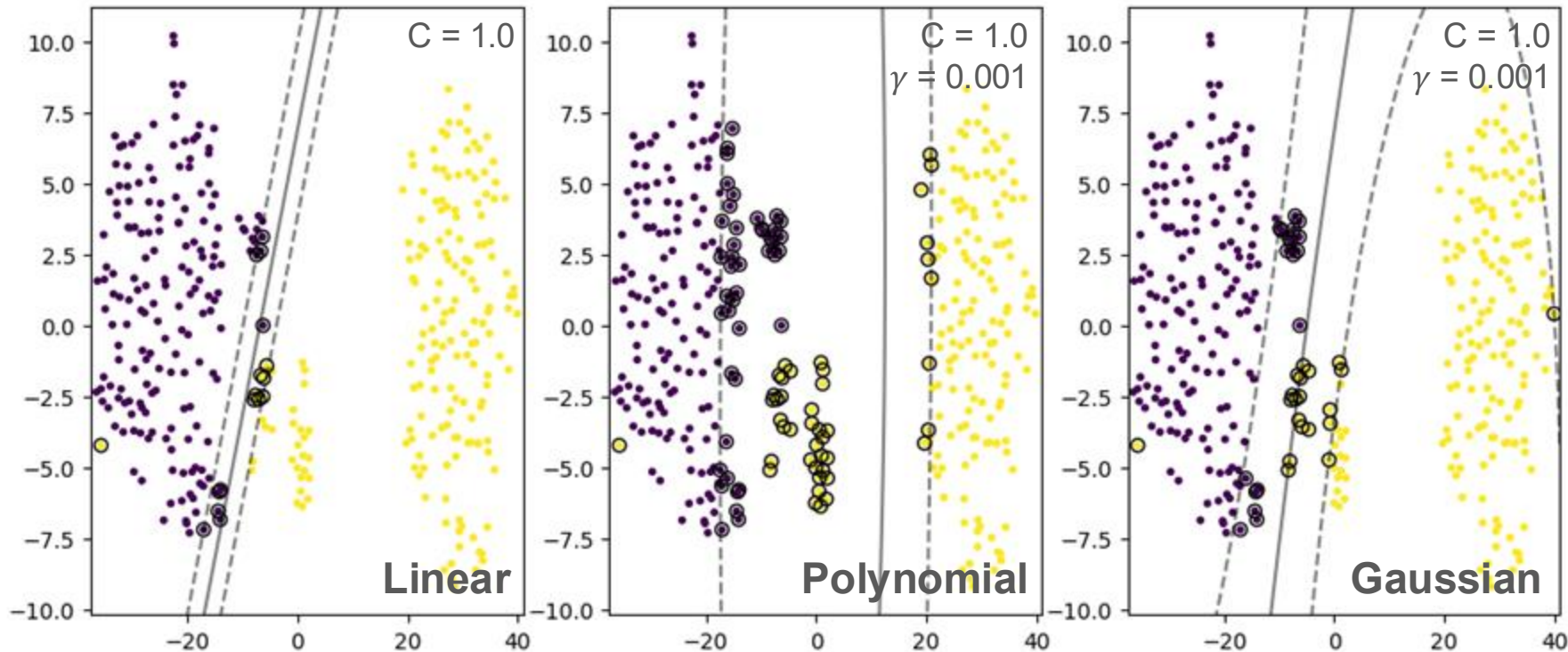
Using different kernels on two separable classes



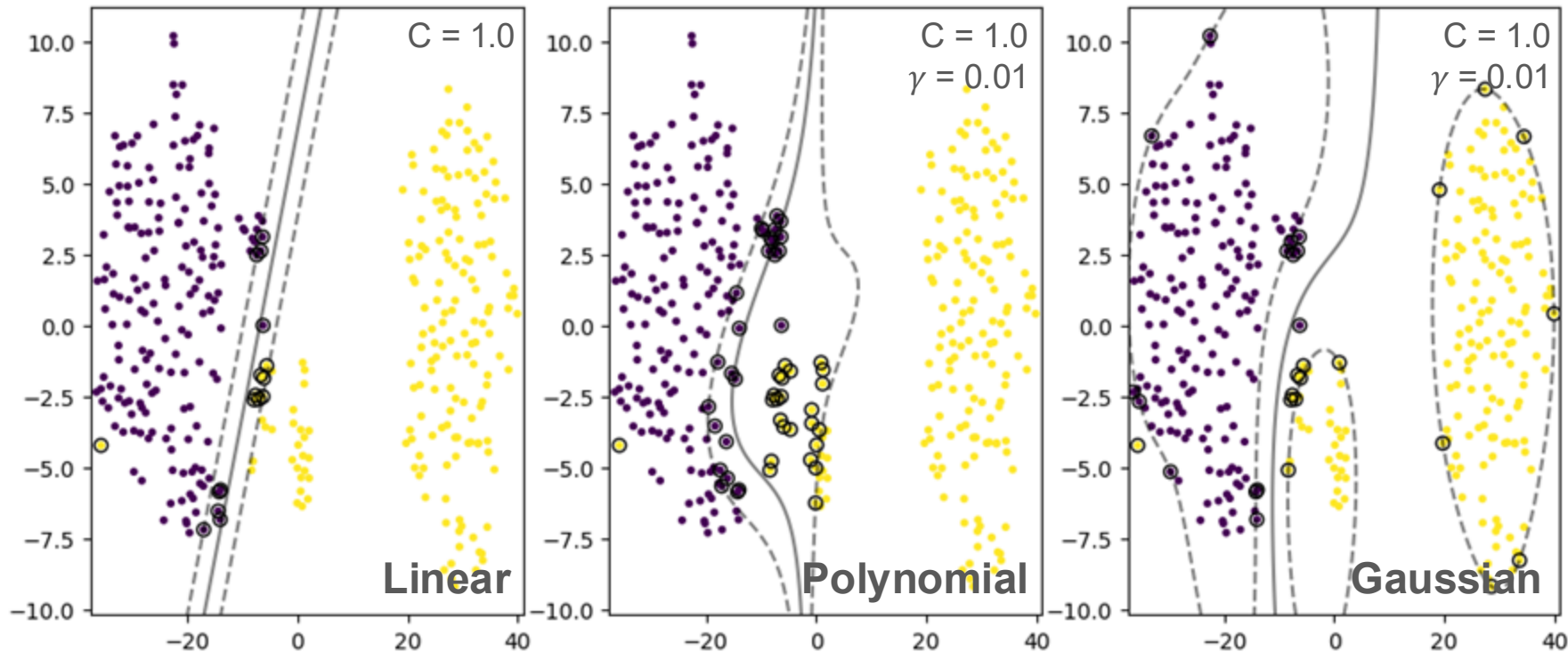
Using different kernels on two non-separable classes



Using different kernels on two non-separable classes



Using different kernels on two non-separable classes



Support Vector Machines with more than two classes

Strategies to implement SVMs with M classes when $M > 2$:

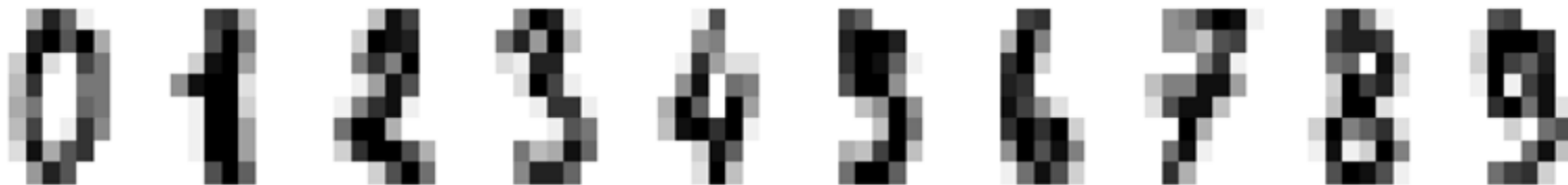
- One-versus-all classification:
 - Learn M binary classifiers (one class versus the other $M-1$)
 - Assign observation $\mathbf{x}_t$ to the class with highest prediction value $g(\mathbf{x}_t) = w_0 + \mathbf{w}^T \mathbf{x}_t$
- All-pairs classification:
 - Learn a binary classifier for each pair of classes, i.e. $\binom{M}{2}$
 - Assign observation $\mathbf{x}_t$ to the class most frequently predicted among the classifiers (majority voting)



Hands-on Coding Session W8S1

The digits dataset

- Number of Instances: 1797
- Number of Attributes: 64
- Attribute Information: 8x8 image of integer pixels in the range 0..16
- Classes: 10 classes where each class refers to a digit



The diabetes dataset

- Number of Instances: 442
- Number of Attributes: 10 numeric predictive values*
age, sex, bmi (body mass index), bp (average blood pressure),
s1_tc (total serum cholesterol), s2_ldl (low-density lipoproteins),
s3_hdl (high-density lipoproteins), s4_tch (total cholesterol / HDL),
s5_ltg (log of serum triglycerides level), s6_glu (blood sugar level)
- Target: a measure of disease progression one year after baseline

* feature variables have been mean centered and scaled by the standard deviation times the square root of “n_samples”

Hands on coding session W8S1



- Get familiar with the notebook for W8S1 and its contents
- Look at the documentation for some of the functions used there from python modules:
 - Scikit-Learn: <https://scikit-learn.org/0.16/index.html#>
- Work on exercises in Part 1: Maximal Margins and Kernels
- Work on exercises in Part 2: Support Vector Classification
- Work on exercises in Part 3: Support Vector Regression

Week 8 - Closing remarks

Wrap-up and Homework 8

- What did we look at today?
- Further reading and support materials
- Seventh homework assignment

What did we look at today?

Maximal Margin and Support Vectors	● Maximal Margin Classifier ● Support Vector based Classification ● Support Vector based Regression
Support Vector Machines	● Kernel Methods (Linear, Polynomial, and Gaussian) ● Classification with more than two classes
Hands-on coding session W8S1	● Maximal Margins and Kernels ● Support Vector Classification ● Support Vector Regression

Further reading and support materials

- James, Witten, Hastie, Tibshirani, Taylor, “An Introduction to Statistical Learning with Applications in Python” (Chapter 9)
<https://www.statlearning.com/>
- Abu-Mostafa, Magdon-Ismael, Lin, “Learning From Data” (Chapter 8) <https://work.caltech.edu/telecourse>

Eight homework assignment (due date 08/24)

- Complete all coding exercises from coding session W8S1





Remember:

Office Hours

Tuesday 08/19 6:00-7:00pm

Next Class

Thursday 08/21, 5:30-9:30pm

Eight Homework Due

Sunday 08/24, 11:59pm

Quiz 3

Thursday 08/21, 6:00-7:00pm